FILM IMAGING

Dr.Manila

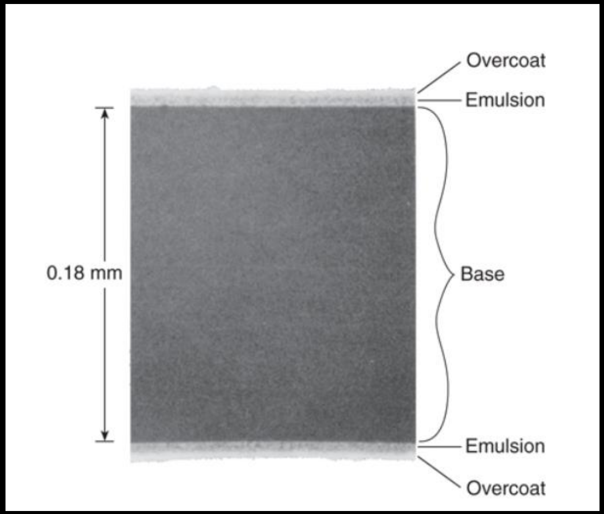
OMFS 553

Radiographic Film Overview

- Single-use image receptor
- Silver halide crystals

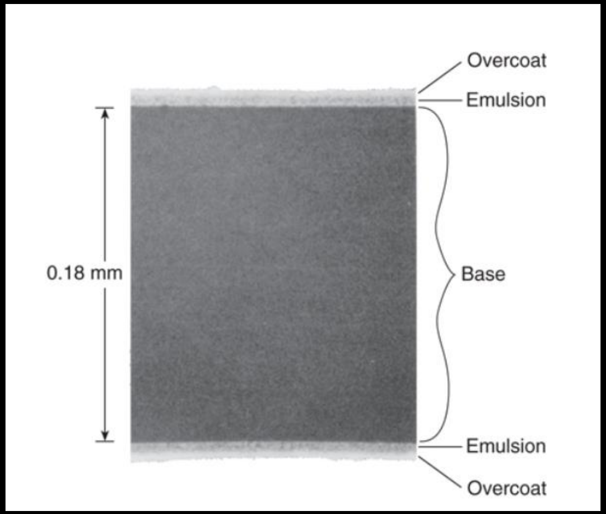
Film Structure

- Base (polyester)
- Emulsion (silver halide, gelatinous matrix)
- Protective layer



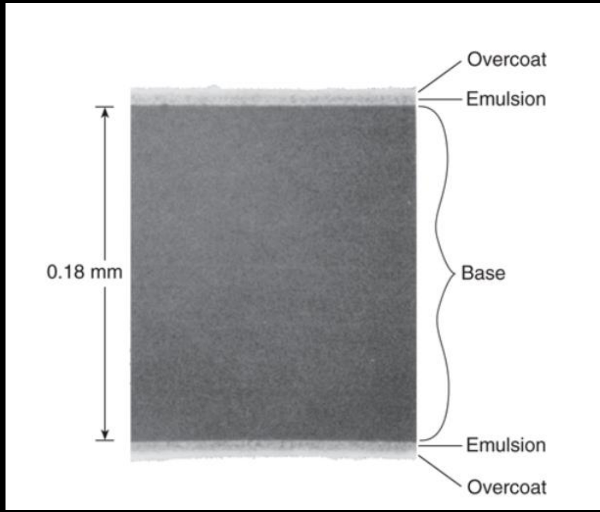
Emulsion

- Silver bromide + iodide
- Light & x-ray sensitive
- Latent image formed here



Film Base

- Polyester (polyester polyethylene terephthalate)
- Flexible & translucent
- No effect on image



Film Packet Components

- Double-emulsion film
- Black paper wrapper
- Lead foil
- Plastic cover

Lead Foil Functions

- Reduces backscatter
- Improves contrast
- Decreases patient dose

Reversed Film Error

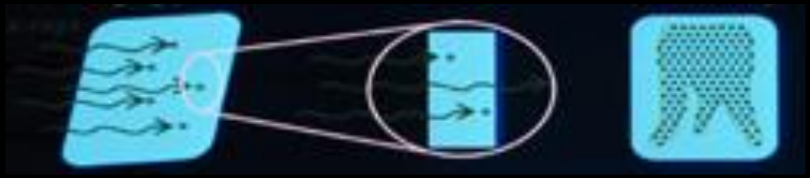
- Light image
- Herringbone pattern
- Right/Left reversed

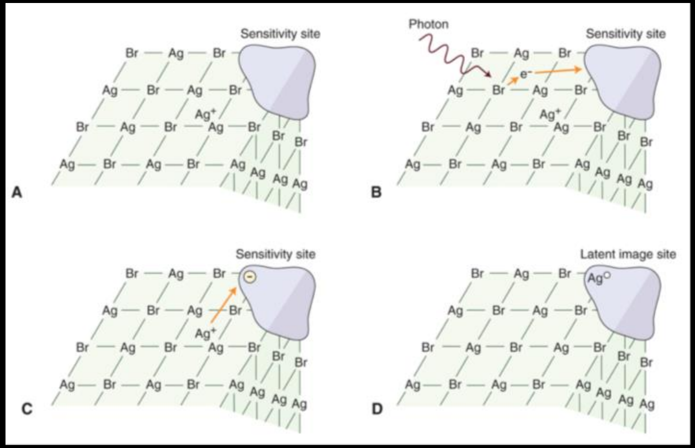
Identification Dot

- Faces x-ray tube
- Guides mounting
- Convex toward viewer

Latent Image

- Invisible image after exposure
- Energized silver halide
- Requires processing





Processing Overview

- Developing
- Fixing
- Drying

Developer Function and Components

- Exposed crystals → metallic silver
- Controls density

- Developing agent
- Accelerator
- Preservative
- Restrainer

Fixer Function and Components

- Removes unexposed crystals
- Hardens emulsion
- Stops development

- Clearing agent (thiosulfate)
- Acidifier
- Preservative
- Hardener

CALIFORNIA NORTHSTATE UNIVERSITY COLLEGE of DENTAL MEDICINE

Components and Function of Radiographic Developer Solution

Component	Ingredient	Function
Developer	Phenidone	Serves as the first electron donor that converts silver ions to metallic silver at the latent image site. This electron transfer generates the oxidized form of phenidone
	Hydroquinone	Hydroquinone provides an electron to reduce the oxidized phenidone back to its original active state so that it can continue to reduce silver halide grains to metallic silver
Activator	Sodium hydroxide or potassium hydroxide	Maintains an alkaline pH (approximately 10) for activity of developer. Causes the gelatin to swell so that the developing agents can diffuse more rapidly into the emulsion to reach silver bromide crystals
Preservative	Sodium sulfite	Antioxidant that extends life of developing solution. Combines with oxidized developer to produce a compound that subsequently stains images brown if not washed out
Restrainer	Bromine-containing compounds	Restrains development of unexposed silver halide crystals. Acts as antifog agents and increases contrast

Components and Function of Radiographic Fixer Solution

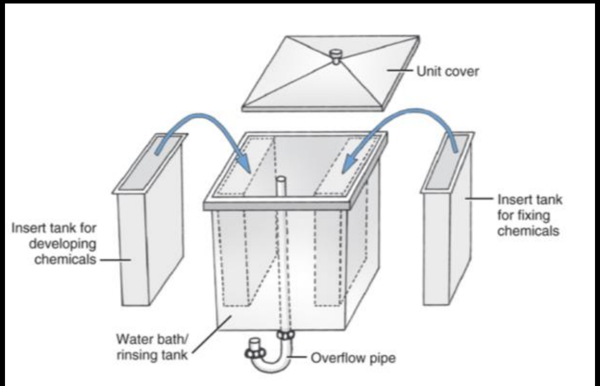
Component	Ingredient	Function
Clearing agent	Ammonium thiosulfate	Dissolves the unexposed silver halide grains Excessive fixation (hours) results in a gradual loss of film density because the grains of silver slowly dissolve in the fixing solution
Acidifier	Acetic acid	Maintain an acidic pH (approximately 4–4.5) Promotes good diffusion of thiosulfate into the emulsion and of silver thiosulfate complex out of the emulsion Inactivates any residual developing agents
Preservative	Ammonium sulfite	Prevents oxidation of ammonium thiosulfate, which is unstable in the acidic environment
Hardener	Aluminum sulfate	Complexes with the gelatin during fixing to prevent damage to the gelatin overcoat during subsequent handling

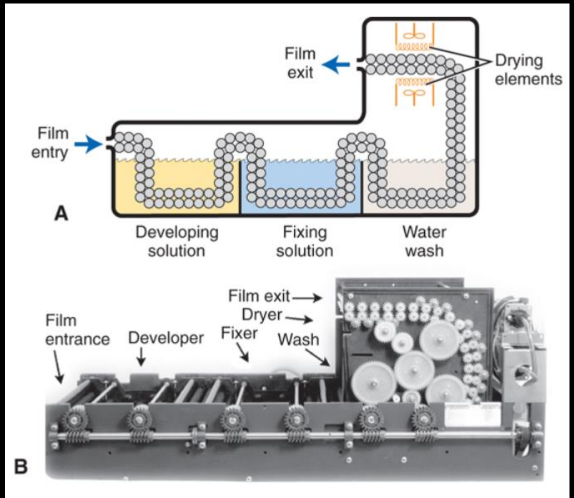
Washing

- Removes chemicals
- Prevents brown stains

Processing Methods

- Manual
- Automatic





Automatic Processing

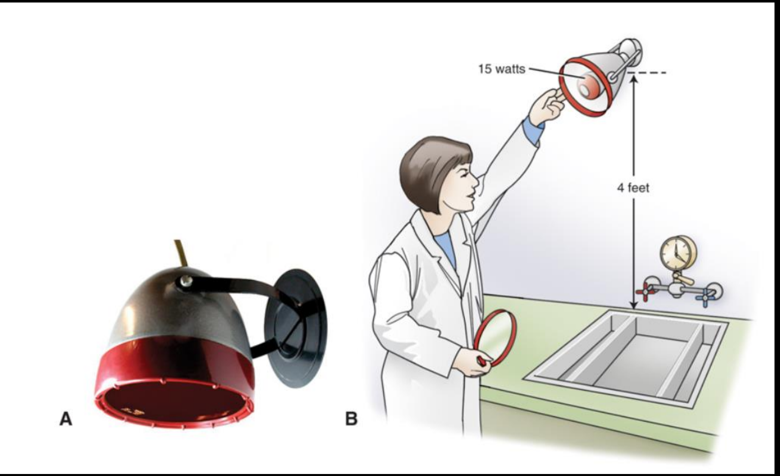
- ~82°F
- Faster
- Special chemicals

Darkroom Requirements

- Light-tight
- Safelight
- Temp/humidity control

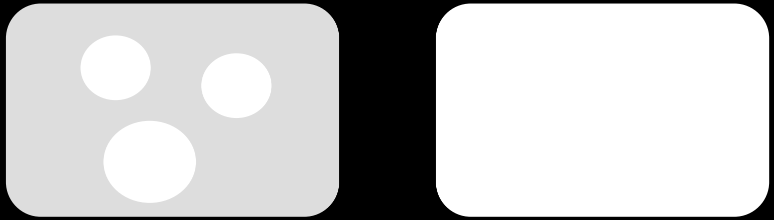
Safelight

- Red or amber filter
- Low wattage
- ≥4 ft from work area



Coin Test

- No coin image = pass
- Coin visible = fail

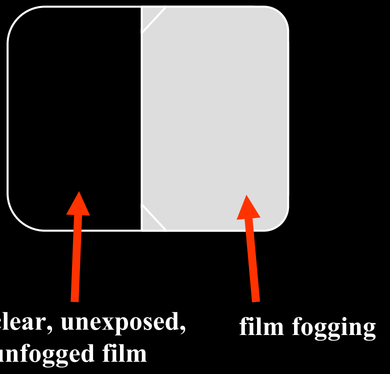


Film Fog

- Uniform gray appearance
- Decreased contrast

Causes

- Light
- Heat
- Chemical contamination
- Expired film



Dark Film Causes

- Developer too hot
- Too much time in developer
- Exposure to light (opening door, turning on light, light leaks around door, incorrect or cracked filters)

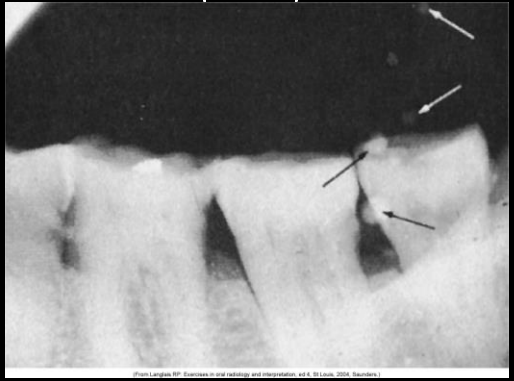
Light Film Causes

- Developer too cold
- Not enough time in developer
- Under replenishment (developer gets weak)
- Contaminated developer
- Excessive fixation

Common Artifacts

- Fingernail
- Fingerprint
- Static
- Scratches

Air bubbles (white)



Fingernail artifact (black)

Scratches

Fingerprint artifact (black)

Static electricity (black branching lines)

Common Artifacts

Fixer spots (light)

Developer spots (dark)

Developer cut-off

Overlapped film

Fixer cut-off